CHAPTER

ATA 70

STANDARD PRACTICES - ENGINE

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Contents

Inspection Procedures

The maintenance procedure that follows is for the inspection of engine components.

Procedure:

Open the access doors and/or lower cowl as necessary for access to do the inspection procedure.

A. General:

(1) To extend the life of the engine and to maintain performance, it is important to carefully do a complete inspection. Make sure that no parts are missing or have become loose. Damage can be caused by the conditions that follow:

- Incorrect clearance
- Incorrect lubrication
- Unwanted movement of parts that are bolted or pressed together
- Too much load
- Load distribution that is not symmetrical
- Heat
- Shock
- Minor damage, such as scratches, tool marks or cracks that have become extended
- Foreign matter, such as grit, chips, moisture or chemicals
- Incorrect removal or installation techniques

B. Inspection Procedures:

(1) Do all of the inspection procedures in an area that has good lighting, is clean and has no dust. Attach suitable tags to all of the parts to identify the necessary repair or replacement. Only a visual inspection is necessary for most parts but you have to use gages or other equipment to measure other parts. Some damage can only be found if you use the magnetic particle or fluorescent penetrant inspection method.

(2) Examine the parts for alignment, distortion, foreign matter, looseness, sharp edges, scratches, taper and wear. Do a check, also, for the following conditions:

- Obstructed pipes, tubes and holes in cases and manifolds
- Contact patterns on gear teeth and splines
- Corrosion on magnesium parts
- Tightness of plugs
- Alignment and protrusion length of studs and dowels
- Completeness of protective surface coatings
- Condition of threads
- Smoothness and flatness of mounting pads and seating surfaces.

NOTE: Use pencil carbon paper to do a smear–type indication of surface smoothness.

(3) Examine all of the external tubes, unless specified differently, for the following conditions:

(a) Acceptable Damage:

1. Fuel tubes exhibiting fretting, corrosion damage, pitting or groups of pitting, nicks and scratches of not more than 0.005 in. (0.127 mm) deep are acceptable. Repair acceptable damage within 600 Flight Hours (FH).

NOTE: Repair of the tube can be done on or off wing.

2. Fuel tubes exhibiting fretting, corrosion damage, pitting or groups of pitting, nicks and scratches of between 0.005 in. (0.127 mm) and 0.010 in. (0.254 mm) in depth on straight sections of the tube only requires replacement within 600 FH. No repair is required.

(b) Dents:

1. Dents that are rounded and have a depth not more than 10% of tube outside diameter and have a length and width at least three times the depth are acceptable.

2. No more than three dents per 12 in. (304.8 mm) length of tube are acceptable. The dents must be separated by 0.250 in. (6.35 mm) minimum.

3. Dents within 1.00 in. (25.4 mm) of a ferrule weld, braze or bond are not acceptable.

Lockwire Alignment

The maintenance procedure that follows is for the alignment of lockwire.

Procedure

Refer Figures 70-1, 70-2 & 70-3

A. General:

(1) After installation, the lockwire must be tight, so that it does not break if there is a rub or vibration.

(2) Install the lockwire so that it tightens to keep a part in the correct position. This prevents a part from becoming loose.

(3) Do not stress the lockwire too much. If you twist the lockwire too tightly, vibration can break it.

(4) Pull the lockwire tight when you twist it, however, it must have minimum tension, if any, when it is installed.

(5) Bend the ends of the lockwire towards the engine or part. If you do not do this, the sharp ends can cause injury to persons.

B. Alignment:

(1) Before you install the lockwire, do the checks that follow:

- Make sure that the bolts are correctly torqued.
- Look to see if the lockwire holes are in the correct position.

(2) Do not tighten the bolt more or less than the limits to get the correct alignment.

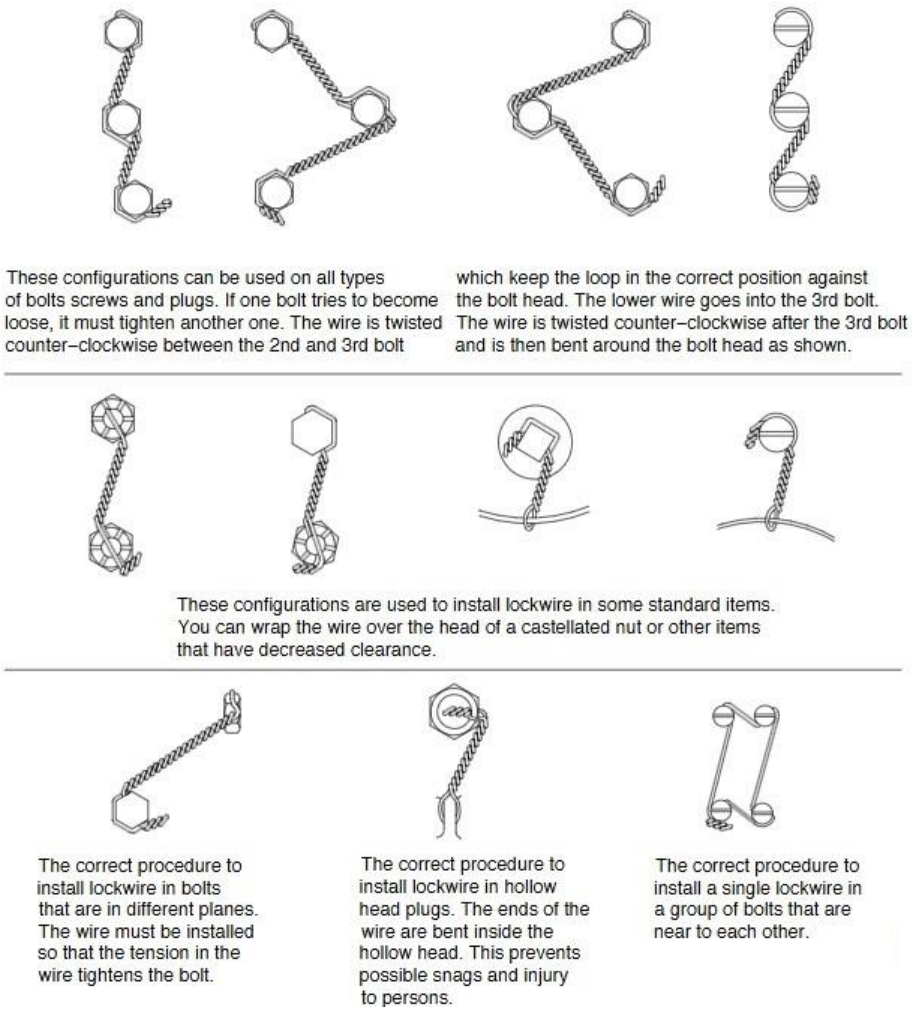


Figure 70-1 Lockwire Configurations and Alignment (Sheet 1 of 3)

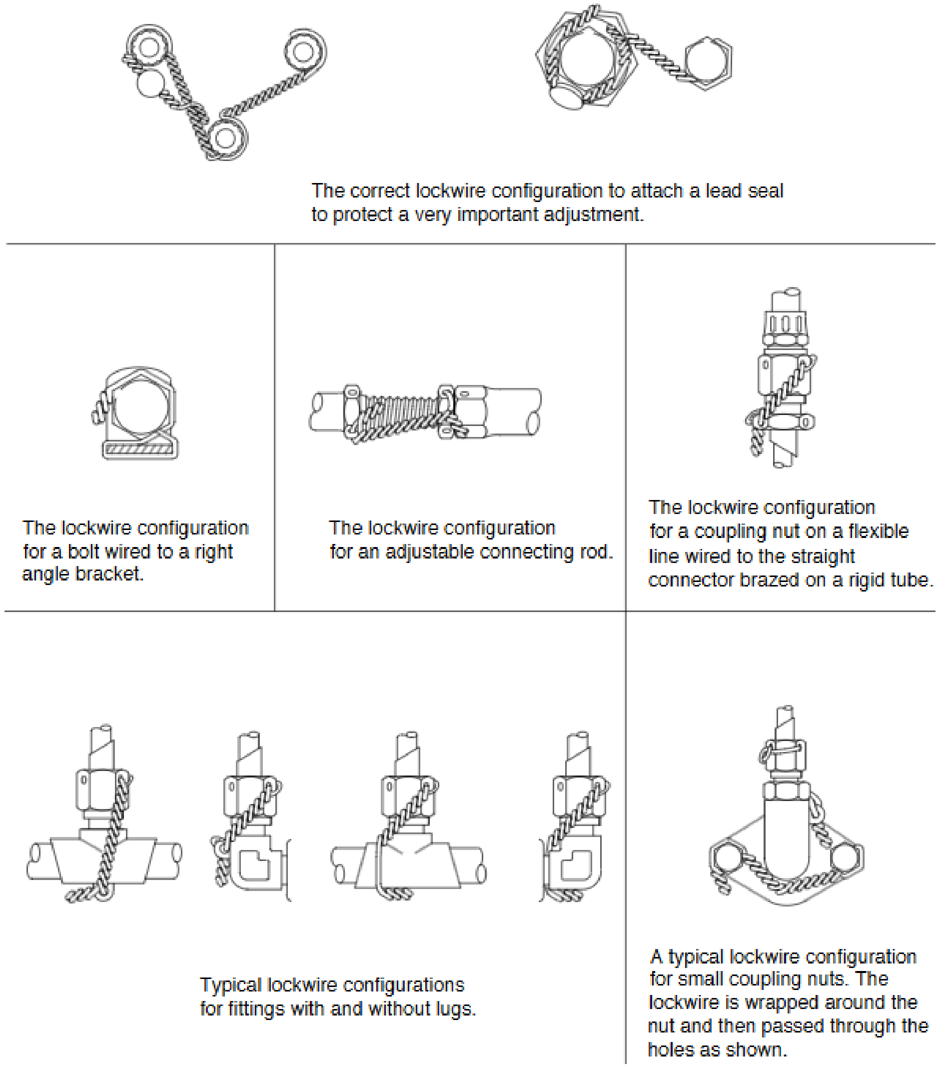


Figure 70-2 Lockwire Configurations and Alignment (Sheet 2 of 3)

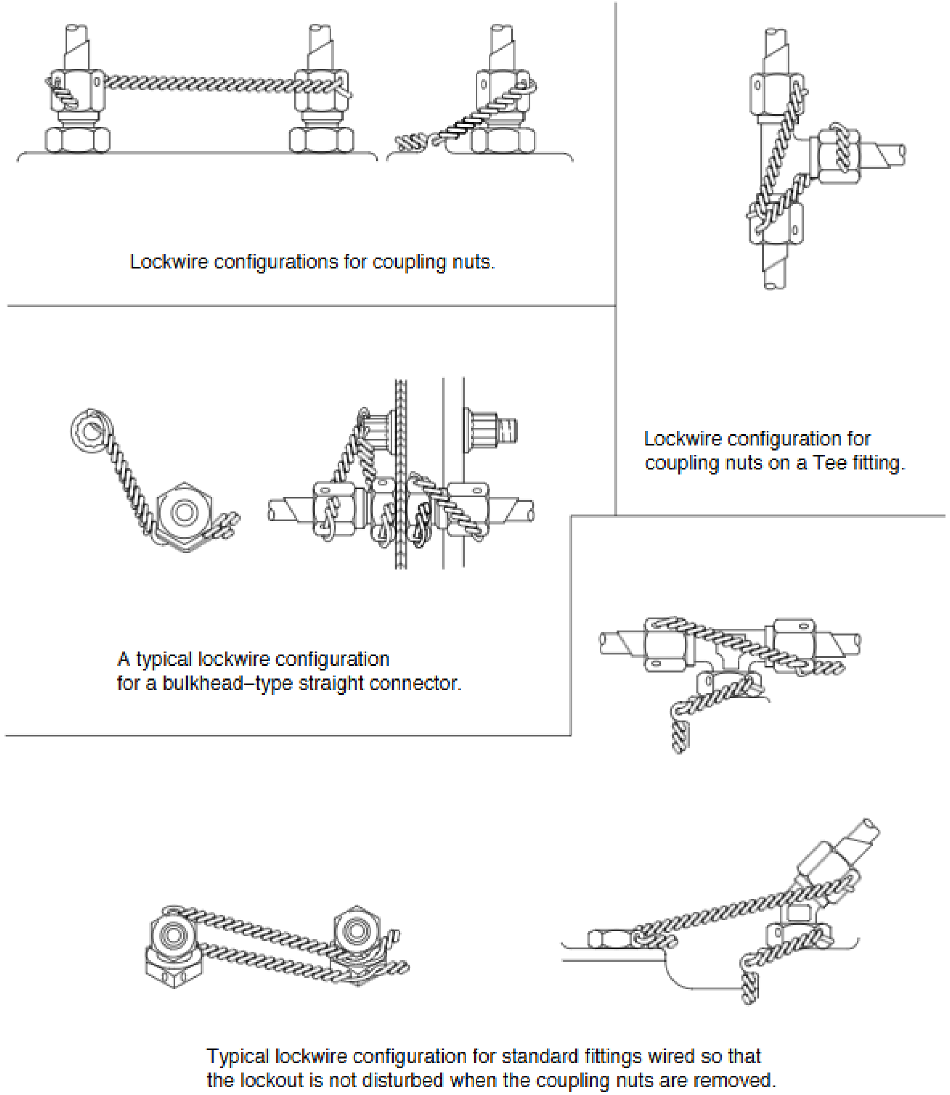


Figure 70-3 Lockwire Configurations and Alignment (Sheet 3 of 3)

Recommended Engine Operation Practices for Harsh Environment

The maintenance procedure that follows is for the recommended engine operational practices for harsh environment.

Procedure

A. For engines exposed to high ambient temperature and to reduce thermal stress on hot section components, P&WC recommends to:

(1) Start using battery cart / APU power at all locations.
(2) To cool the engine, Motor engine 10 −15 seconds prior to selecting CLA at start/feather position if engine temperature before starting is above 200°C. Ensure to obey starter limitation.
(3) Cool down engines before shutdown.
(4) Use reduced take off power where possible.
(5) Transition from Take-off power setting to climb setting as soon as practical taking into consideration obstacles clearance and minimum climb rate required.
(6) Operate at reduced Climb and Cruise power where possible.
(7) Select Cabin bleed “OFF” during take-off.
(8) Avoid use of reverse on landing (except for emergency).
(9) Monitor Bleed valve operation based on ECTM results.
(10) When airframe bleed leaks are suspected perform Power Assurance Checks with bleed ports capped and uncapped.
(11) Check the airframe bleed air system and ducting for leaks on a regular basis.

B. For engines exposed to sand erosion and FOD, P&WC recommends to:
(1) Restrict CLA movement when aircraft is stationary
(2) Taxi using both engines (both engines out of feather and move PLA equally)
(3) Always use intake blanking covers when the AC is parked overnight
(4) Avoid use of reverse on landing (except for emergency)
(5) Avoid running into sand wakes after landing and prior to takeoff
(6) Move propeller out of feather when engine has stabilized
(7) Operate from paved runway & paved ramp

C. For engines exposed to moderate to severe marine environment and/or volcanic fumes, P&WC recommends to:

(1) Perform compressor/turbine wash every 50 FH. Adjust washing interval based on level of exposure to the salt laden atmosphere
(2) Perform an initial borescope inspection of the HP, LP and PT sections at 2000 FH TSN/TSO for evidence of sulphidation attack on turbine blades. Adjust inspection interval based on inspection results
(3) Perform an external engines wash every 1200 FH to reduce the accumulation of salt deposits on the engine. Adjust wash frequency based on inspection results
(4) Perform an initial visual Inspection of all magnesium parts at 1200 hrs TSN/TSO

- Inspect external surfaces and air inlet gas path components for the presence of corrosion and/or any surface damages
- Do the procedure to repair magnesium surfaces in accordance with the Dash8 AMM. Apply 2 to 3 coats of paint to the affected area
- Repeat the repair procedure as often as required to maintain painted surface in good condition on all exposed surfaces
- Pay particular attention to bolted flanges or areas where the painted surface may be compromised.

Removal/Installation of Nipples

The maintenance procedure that follows is for the removal / installation of nipples.

- Make sure that the aircraft is in the standard configuration for maintenance.
- Obey all the electrical/electronic safety precautions
- Obey all the electrostatic discharge safety precautions

Procedure

Refer Figure 70-4

A. Replace the nipple (1) as follows:

(1) Remove the lockwire, if applicable.

(2) Remove the nipple (1).

(3) Remove and discard the packing (2).

(4) Lubricate the threads of the replaced nipple (1) with the engine oil (03–06).

(5) Lubricate the packing (2) with the engine oil (03–06).

(6) Install the packing (2) in the groove of the nipple (1).

(7) Install the nipple (1) in the boss (3).

(8) Torque the nipple (1) to the correct value shown in related installation procedure.

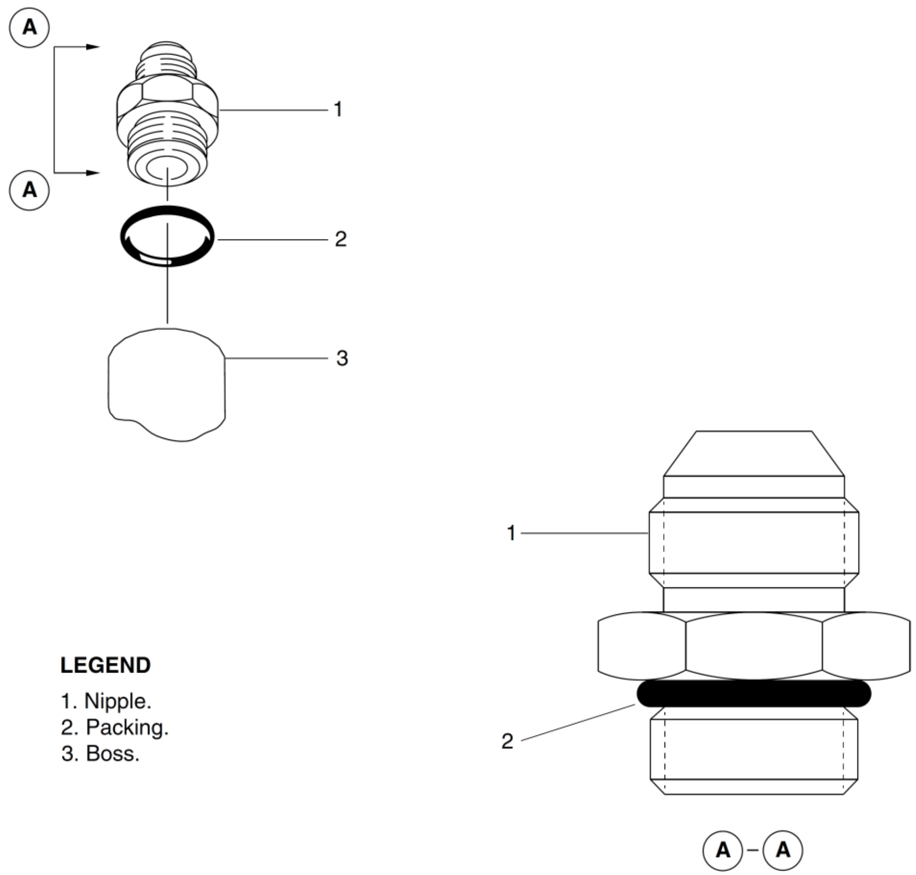


Figure 70-4 Removal/Installation of Nipples

Removal/Installation of Elbow and Tee Assemblies

The maintenance procedure that follows is for the removal/installation of elbow and tee fittings.

- Obey all the electrical/electronic safety precautions
- Obey all the electrostatic discharge safety precautions

Procedure

Refer Figure 70-5

A. Removal:

(1) With a marker (13−22), mark the angular location of the elbow/tee fitting.

(2) Remove the lockwire, if applicable.

(3) Loosen the nut (2) and remove the elbow/tee fitting (1).

(4) Remove and discard the packing (4) and the backup ring (3).

(5) Remove the nut (2).

B. Installation:

(1) Lubricate the threads of the elbow/tee fitting (1) with engine oil (03−06).

(2) Fully install the nut (2).

(3) Lubricate the backup ring (3) with engine oil (03−06) and install it on the elbow/tee. Push the backup ring fully into the counterbore of the nut (2).

(4) Lubricate the packing (4) with engine oil (03−06) and install it in the groove of the elbow/tee fitting.

(5) Turn the nut (2) until the backup ring (3) is in the groove of the elbow/tee fitting (1) and the packing (4) is against the first thread.

(6) Install the elbow/tee fitting (1) in the boss (5). Let the nut (2) turn with the elbow/tee fitting (1) until the packing (4) touches the mating face of the boss (5). You will feel an increase in the torque at this position.

(7) Hold the nut (2) so that it does not turn while you turn the elbow/tee fitting (1) into the boss 1½ turns more.

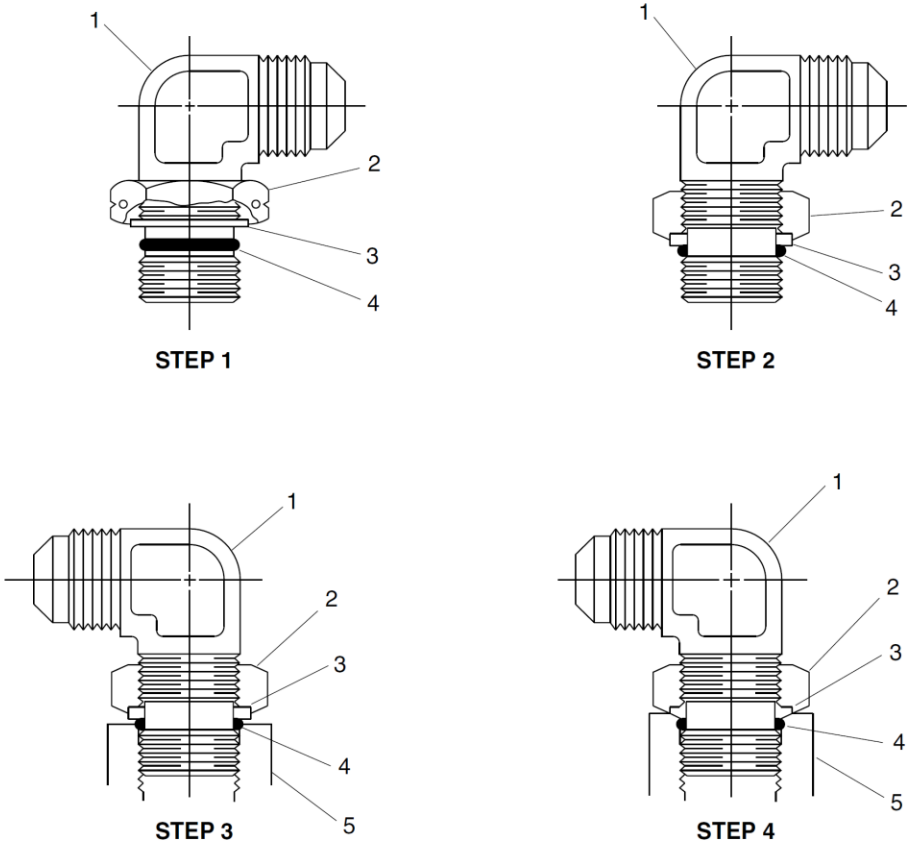
NOTE: If necessary, you can turn the elbow/tee fitting up to one turn more to correctly align it.

(8) While you hold the elbow/tee fitting (1) in the correct position, tighten the nut (2) and then torque it to the correct value shown in the relevant installation procedure.

NOTE: With the elbow/tee correctly aligned, the nut must touch the surface of the boss without going more than the recommended torque. There must be no extrusion of the packing or the backup ring.

(9) Install the lockwire as necessary.

C. Remove all tools, equipment, and unwanted materials from the work area.



LEGEND

1. Fitting.
2. Nut.
3. Backup ring.
4. Packing.
5. Boss.

Figure 70-5 Elbow and Tee Assemblies – Removal/Installation

Removal/Installation of Lockwire

The maintenance procedure that follows is for the removal/installation of lockwire.

Procedure

Refer Figure 70-6

A. General:

(1) Lockwire must be tight, after installation, so that it does not break if there is vibration.

(2) Install the lockwire so that it tightens to keep a part in the correct position. This prevents a part from becoming loose.

(3) Do not stress the lockwire too much. Vibration can break the lockwire if you twist it too tightly.

(4) Pull the lockwire tight when you twist it, however, it must have minimum tension, if any, when it is installed.

(5) Bend the ends of the lockwire towards the engine or part. If you do not do this, the sharp ends can cause injury to persons.

B. Removal:

(1) Do not twist off the lockwire with pliers. This can cause damage to the related components.

(2) Cut carefully the lockwire near to the component locking hole.

C. Installation:

(1) Hold the lockwire at the ends or at a location that will not be twisted. Do this so that you will cause damage to the twisted section.

(2) Be careful not to cause nicks, kinks or other damage to the lockwire.

(3) When you cut off the ends, leave a minimum of three complete turns after the loop that safeties the component. Make sure that the ends which you cut off do not fall into the engine.

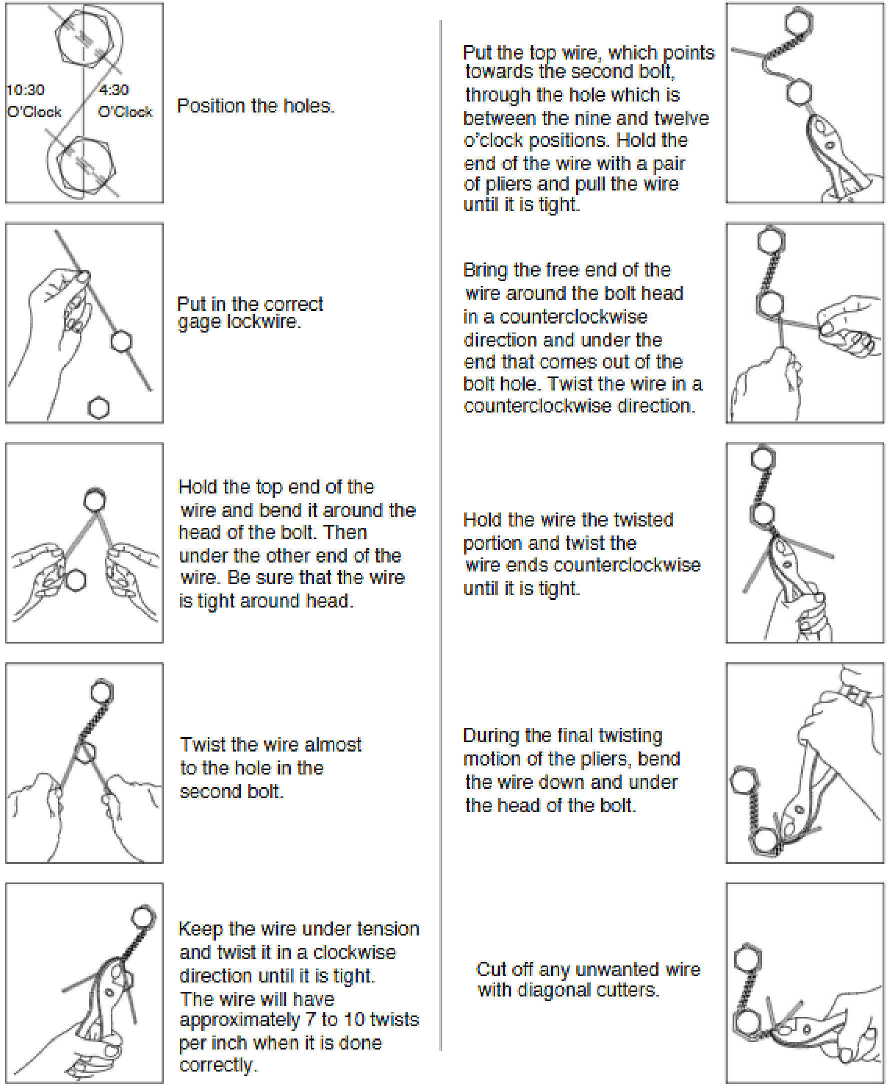


Figure 70-6 Lockwire (Typical) – Installation

Locking practices - Keywashers, Key–Type

The maintenance procedure that follows is for installation of key–type keywashers (written "keywasher" in this procedure).

Procedure

Refer Figure 70-7

A. General:

(1) Only use a keywasher once. Discard any keywasher that has been used.

(2) When you bend the keys, do not use a tool which has a sharp point. This can cause a key to break which can damage the engine.

B. Install keywashers as follows:

(1) Put the keywasher on the bolt or stud with the pre–bent key in the locking hole against the side to prevent movement of the fastener that is being locked.

(2) Bend a minimum of one key against the head of the nut or bolt. A minimum of 75% of the width of the key (measured at the base) must engage the flat of the fastener. You can bend other keys in any applicable way.

(3) Do not use impact–type tools to bend keywashers that are used on fuel nozzles.

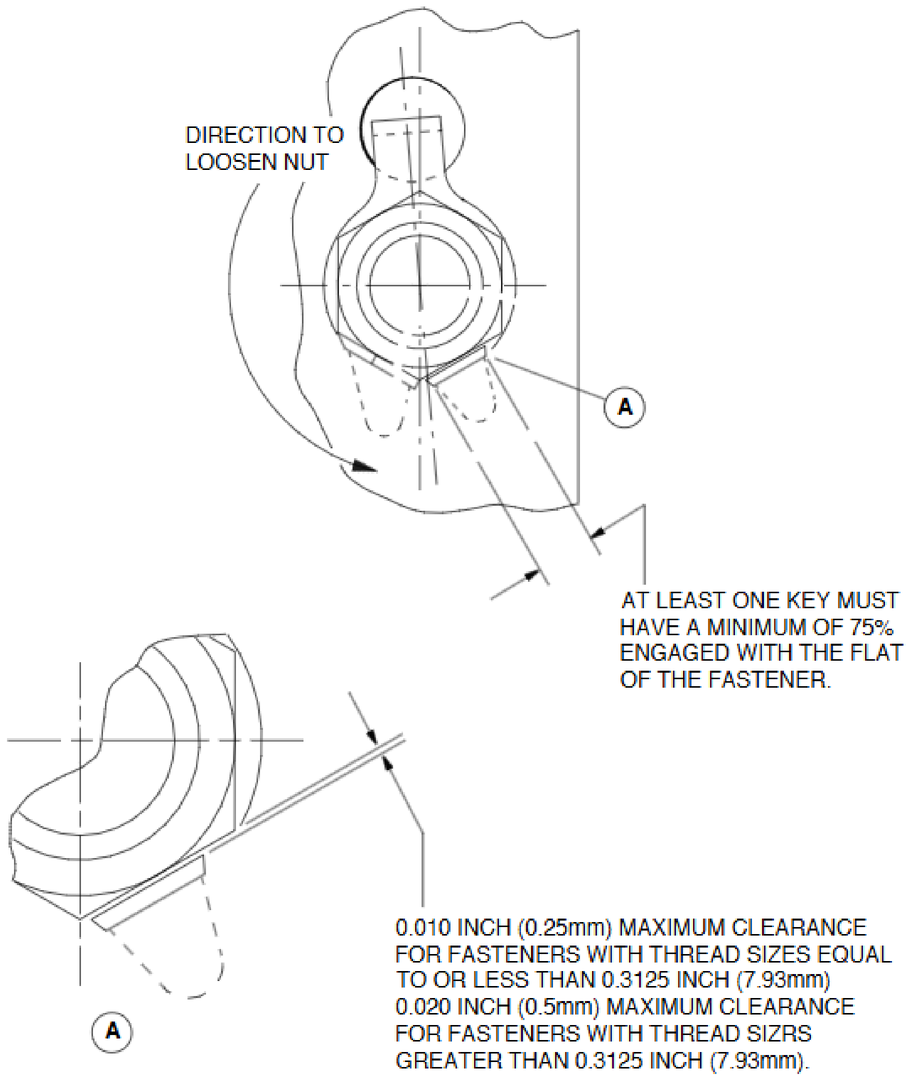


Figure 70-7 Keywashers - Installation

Locking Practices, Keywashers, Cup−Type

The maintenance procedure that follows is for installation of cup−type keywashers (written "cupwasher" in this procedure).

Procedure

Refer Figure 70-8

A. General:

(1) Only use a cupwasher once. Discard any cupwasher that has been used.

(2) When you bend the keys, do not use a tool which has a sharp point. This can cause a key to break off which can damage the engine.

B. Install cupwashers as follows:

(1) Put the cupwasher over the thread with the pre−bent key in the locking slot or groove.

(2) With a suitable crayon, make a mark on the cupwasher and on the adjacent surface.

(3) Tighten the nut to the correct torque.

(4) Visually check the marks that you made before to see if the cupwasher has moved.

(5) If the cupwasher has moved, replace it with a new one and the torquing procedure again.

(6) Bend the OD of the cupwasher into a minimum of two slots of the nut.

(7) The clearance between the OD of the nut and the ID of the cupwasher must not be more than 0.005 in (0.12 mm).

(8) With one of the tools that follow, crimp the cupwasher:

- Drift−type tool. This tool consists of a ring with prongs on the ID. When the tool is put over the cupwasher and moved axially, the tool forms indentations on the OD of the cupwasher.
- Squeeze−action tool. This tool forms indentations by pressing the OD of the cupwasher into the slots of the nut.
- Punch. The impact on this tool forms indentations in the OD of the cupwasher.

(9) The part of the tool that forms the indentations must have a spherical shape. The spherical radius must not be less than 0.050 in (1.27 mm). The radius must not be more than is necessary to form fully the indentation into the applicable slot.

(10) To prevent the internal keys from shearing, the position of the cupwasher must not change when you torque the nut. Use an applicable marking procedure on the cupwasher and an adjacent surface, so that you can see any movement.

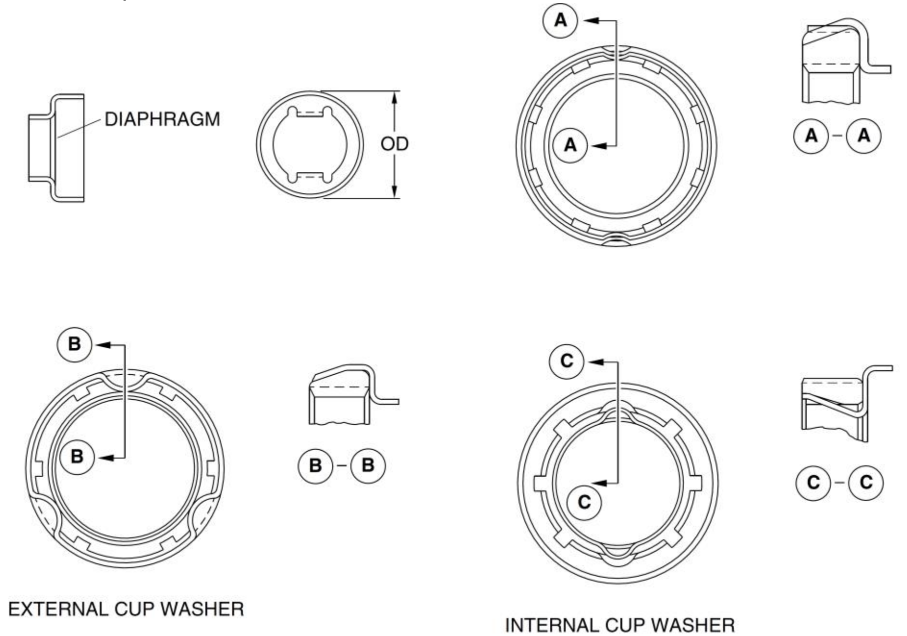


Figure 70-8 Cupwashers – Removal/Installation

Standard Torque for Engines

The maintenance procedure that follows is to torque nut or bolts

Procedure

A. The torque limits are given as follows:

(1) Torque values are at room temperature.

(2) Angles of turn are in degrees.

(3) The torques that are given are the torques which must be applied to the part. They do not take into consideration any adapter or extension used when the part is torqued.

B. You must lubricate the threads with engine oil (03−06), unless stated differently.

NOTE: Nuts or bolts that are silver−plated do not require lubrication before assembly.

C. Torque the nuts or bolts that are on a flange in a star−shaped pattern.

D. Parts that are heated or cooled before assembly must be at room temperature when you apply the final torque.

E. Apply the torque slowly and evenly for more precision

General Torque Recommendation – Castellated Nut

The maintenance procedure that follows is for installation of castellated nuts.

Procedure

A. Install castellated nuts as follows:

(1) Tighten the nut to the minimum recommended torque.

(2) If a locking slot is not aligned with the lockwire or cotterpin hole, then increase the torque until you get the correct alignment. If the necessary torque was more than the limit, then loosen the nut 180 degrees and do the procedure again.

(3) Replace the nut if you cannot get the correct alignment without going more than the torque limit.

General Torque Recommendation – Oil Lubricated Parts

The maintenance procedure that follows is for oil lubricated parts.

Procedure

A. General:

(1) The torque limits that are given in the assembly instructions for oil lubricated parts are correct only if engine oil (03−06) is used on the parts.

B. Lubrication of parts:

(1) Lubricate with a small amount of engine oil (03−06), the threads and the faces of stacked components.

(2) Lubricate with a small amount of engine oil (03−06), the diameters of oil seals before assembly or installation.

General Torque Recommendation – Self–Locking Nuts and Helical Coil Inserts

The maintenance procedure that follows is for self–locking nuts and helical coil inserts.

Procedure

Refer Figure 70-9

A. General:

(1) The locking torque for self–locking nuts and helical coil inserts must be checked at assembly. The locking torque, unless stated differently, must not be less than the values shown in Table 70-1.

(2) Make sure that the fastener is not seated when you check the self–locking torque. This makes sure that only the torque required to overcome the friction holding the thread is measured.

(3) A minimum thread protrusion is necessary to make sure that the locking feature of the self–locking thread engages fully with the fully–formed mating thread and is given in Table 70-1.

Thread size	Torque lbf-in (Nm)	Minimum Protrusion in (mm)
0.1120–40	0.5 (0.06)	
0.1120–48	0.5 (0.06)	
0.1150–40	1.0 (0.11)	
0.1380–32	1.0 (0.11)	0.050 (1.27)
0.1380–40	1.0 (0.11)	0.040 (1.02)
0.1640–32	1.5 (0.17)	0.050 (1.27)
0.1640–36	1.5 (0.17)	0.040 (1.02)
0.1900–24	2.0 (0.23)	0.060 (1.52)
0.1900–32	2.0 (0.23)	0.050 (1.27)
0.2500–20	4.5 (0.51)	0.060 (1.52)
0.2500–28	3.5 (0.40)	0.050 (1.27)
0.3125–18	7.5 (0.85)	0.070 (1.78)
0.3125–24	6.5 (0.73)	0.060 (1.52)
0.3750–16	12.0 (1.36)	0.070 (1.78)
0.3750–24	9.5 (1.07)	0.060 (1.52)
0.4375–14	16.5 (1.86)	0.080 (2.03)
0.3750–20	14.0 (1.58)	0.060 (1.52)
0.5000–14	24.0 (2.71)	0.080 (2.03)
0.5000–20	18.0 (2.03)	0.060 (1.52)
0.5625–12	30.0 (3.39)	0.080 (2.03)
0.5625–18	24.0 (2.71)	0.070 (1.78)
0.6250–11	40.0 (4.52)	0.090 (2.29)
0.6250–18	32.0 (3.61)	0.070 (1.78)
0.7500–10	60.0 (6.78)	
0.7500–16	50.0 (5.65)	
0.8750–9	82.0 (9.27)	
0.8750–14	70.0 (7.91)	
1.0000–8	110.0 (12.43)	
1.0000–14	92.0 (10.40)	
1.1250–8	137.0 (15.48)	
1.1250–12	117.0 (13.22)	
1.2500–12	143.0 (16.16)	

Table 70-1 Minimum Thread Locking Torque and Thread Protrusion

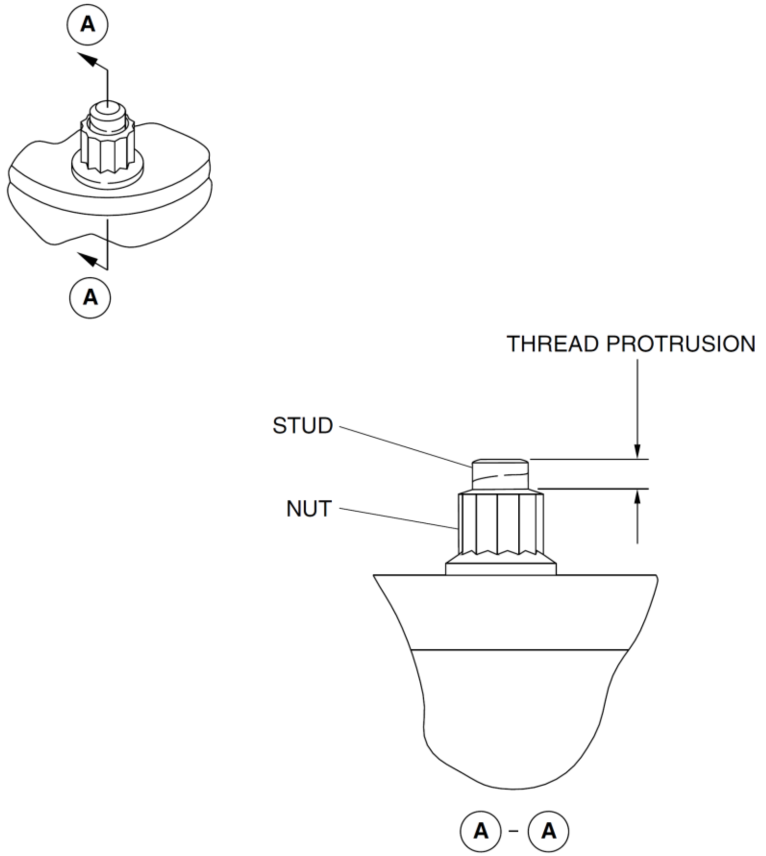


Figure 70-9 Thread Protrusion

Standard Torque Wrenches and Extensions

The maintenance procedure that follows is for the correct use of torque wrenches.

Procedure

A. General:

(1) Do a calibration check of the torque wrenches before you use them. Do not check one torque wrench against another one because you can get results that are not accurate. Some torque wrenches are sensitive to the way they are supported when they are used to torque a part. Always use the instructions that are supplied by the manufacturer.

B. Standard Torque Wrenches and Extensions:

(1) Use the formula that follows to correct the indicated torque when you use an extension or an adapter which has the effect of increasing the length of the torque wrench (*Refer Figure 70-10*, *Configuration A*):

R = bT/a+b = bT/c

a – The effective length of the extension or adapter

- b – The effective length of the torque wrench
- c – The distance through which the force is applied
- R – The reading on the scale or dial of the torque wrench
- T – The required torque on the part

(2) Example:

(a) Required torque (T) = 1440 lbf in (162.7 Nm)

Adapter or extension length = 3 in (76.2 mm)

Torque wrench length = 15 in (381 mm)

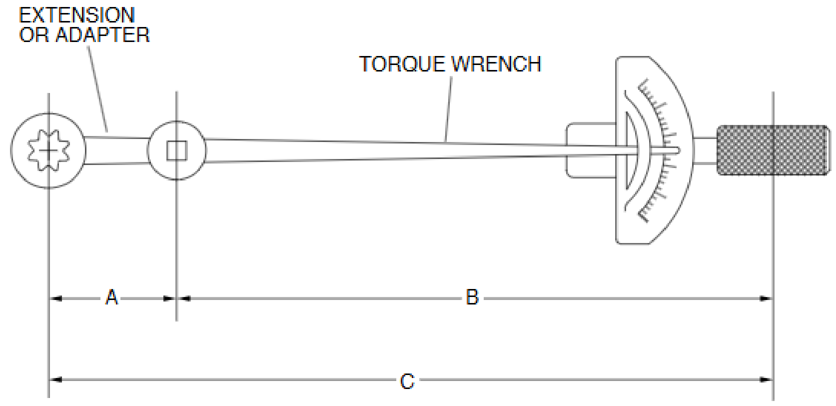
Then:

R (lbf in) = (bT) / (a + b) = (15 x 1440) / (3 + 15) = 1200 lbf in

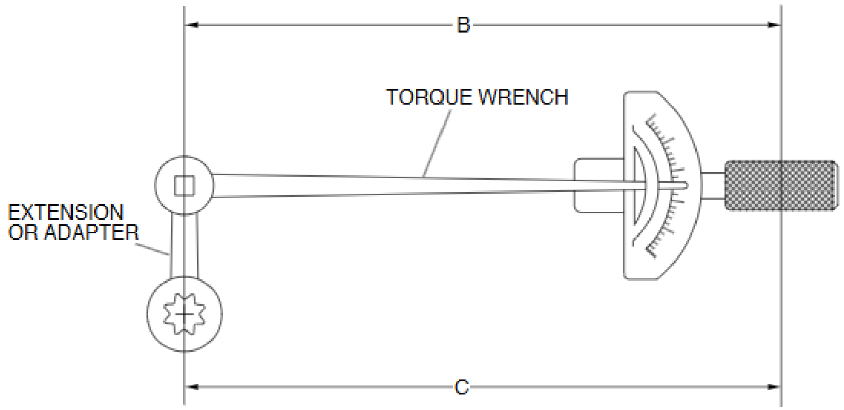
(OR)

R (Nm) = (381 x 162.7) / (76.2 + 381) = 135.58 Nm

(3) You do not need to correct the indicated torque when you use an extension or an adapter which does not change the effective length of the torque wrench (*Refer Figure 70-10*, *Configuration B*).



CONFIGURATION A



CONFIGURATION B

Figure 70-10 Torque Wrench and Extension/Adapter